Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Randomized, Phase 2 Study of Pembrolizumab And Chemotherapy With or Without MK-4830 as Neoadjuvant Treatment for High-Grade Serous Ovarian Cancer **Short Title/Acronym:** Pembrolizumab With Chemotherapy and MK-4830 for Treating Participants With Ovarian Cancer (MK-4830-002) **Protocol Number:** MK-4830-002 **NCT Number:** NCT05446870 **Sponsor Name:** Merck **Investigational Product:** Pembrolizumab, MK-4830, and Chemotherapy (Paclitaxel/Docetaxel, Carboplatin) **Phase of Development:** Phase 2 **Medical Monitor:** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate in participants with high-grade serous ovarian cancer (HGSOC), whether the reduction from baseline in circulating tumor deoxyribonucleic acid (ctDNA) at Cycle 3 (ΔctDNA) is larger in participants receiving MK-4830 + pembrolizumab in combination with standard of care (SOC) therapy than in those receiving pembrolizumab + SOC therapy. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Overall Survival (OS), Objective Response Rate (ORR), complete resection rate during interval debulking surgery, the association of change from baseline in ctDNA with Pathological Complete Response (pCR) and Chemotherapy Response Score (CRS), and the safety and tolerability of the treatment regimens.

2.2 Background & Rationale To address the need for improved neoadjuvant therapies in patients with High-Grade Serous Ovarian Carcinoma and Ovarian Carcinoma. While standard chemotherapy (carboplatin and paclitaxel) is effective, recurrence is common. This study aims to determine whether combining pembrolizumab (an immune checkpoint inhibitor) with MK-4830 (a novel ILT4-targeted monoclonal antibody) and chemotherapy can enhance the immune response against the tumor, shrink the cancer prior to interval debulking surgery, and more effectively reduce circulating tumor DNA in the blood.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab + Chemotherapy) **Blinding:** Open-label **Randomization:** Randomized (1:1 ratio) **Trial Status:** COMPLETED

3.2 Planned Enrollment & Duration Target N: 160 evaluable patients **Number of Sites:** International, multi-center trial **Estimated Study Start:** Mid-2022 **Study Status:** Completed

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Female, age ≥ 18 years. 2. Has histologically-confirmed International Federation of Gynecology and Obstetrics (FIGO) Stage III or Stage IV High-Grade Serous Ovarian Cancer (HGSOC), primary peritoneal cancer, or fallopian tube cancer. 3. Is a candidate for carboplatin and paclitaxel chemotherapy (administered in the neoadjuvant and adjuvant setting). 4. Is a candidate for interval debulking surgery. 5. Is able to provide archival tissue or newly obtained core, incisional, or excisional biopsy of a tumor lesion. 6. Has adequate organ functions.

4.2 Exclusion Criteria 1. Has a non-HGSOC histology. 2. Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 3. Has a known additional malignancy that is progressing or has required active treatment within the past 3 years. 4. Has received prior treatment for any stage of ovarian cancer (OC), including radiation or systemic anticancer therapy. 5. Planned or has been administered intraperitoneal chemotherapy as first-line therapy. 6. Has received prior therapy with an anti-programmed cell death 1 protein (PD-1), anti-programmed cell death 1 ligand 1 (PD-L1), anti-programmed cell death 1 ligand 2 (PD-L2), anti-immunoglobulin-like transcript 4 (ILT4), or anti-human leukocyte antigen (HLA)-G agent or with an agent directed to another stimulatory or coinhibitory T-cell receptor. 7. Has received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 8. Is currently participating in or has participated in a study of an investigational agent or has used an investigational device within 4 weeks before the first dose of study intervention. 9. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy (in dosing exceeding 10 mg daily of prednisone equivalent) or any other form of immunosuppressive therapy within 7 days prior to the first dose of study medication. 10. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 11. Has severe hypersensitivity to pembrolizumab, carboplatin, paclitaxel (or docetaxel, if applicable), Avastin or biosimilar (if using) and/or any of their excipients. 12. Has an active autoimmune disease that has required systemic treatment in the past 2 years. 13. Has an active infection requiring systemic therapy. 14. Has a known history of human immunodeficiency virus (HIV) infection. 15. Has a known history of hepatitis B or known active hepatitis C virus infection. 16. Has received colony-stimulating factors within 4 weeks prior to receiving study intervention on Day 1 of Cycle 1. 17. Has had surgery <6 months prior to Screening to treat borderline ovarian tumors, early-stage OC, or early-stage fallopian tube cancer. 18. Has a known psychiatric or substance abuse disorder that would interfere with the participant's ability to cooperate with the requirements of the study. 19. Has current, clinically relevant bowel obstruction. 20. Has a history of hemorrhage, hemoptysis, or active gastrointestinal (GI) bleeding within 6 months prior to randomization. 21. Has uncontrolled hypertension. 22. Has had an allogenic tissue/solid organ transplant. 23. Has either had major surgery within 3 weeks of randomization or has not recovered from any effects of any major surgery.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Neoadjuvant Phase (Before Surgery):** Pembrolizumab 200 mg + Paclitaxel 175 mg/m^2 (or Docetaxel 75 mg/m^2) + Carboplatin AUC 5 to 6, with or without MK-4830 800 mg. Administered on Day 1 of each 21-day cycle (Q3W) for 3 cycles. * **Adjuvant Phase (After Surgery):** Pembrolizumab 200 mg + Paclitaxel 175 mg/m^2 (or Docetaxel 75 mg/m^2) + Carboplatin AUC 5 to 6, with or without MK-4830 800 mg (Avastin/bevacizumab is permitted at the investigator's discretion). Administered on Day 1 of each 21-day cycle (Q3W) for 3 cycles. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Total of 6 cycles (3 neoadjuvant + 3 adjuvant), equivalent to 18 weeks of active systemic therapy.

5.2 Concomitant & Prohibited Therapy Permitted: Avastin (bevacizumab) or biosimilars during the post-surgery adjuvant phase per institutional guidelines. Standard medications for symptom management (e.g., anti-emetics). **Prohibited:** Use of concurrent investigational agents or live/live-attenuated vaccines within the specified windows prior to or during the active treatment period.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy, Baseline ctDNA Labs
Neoadjuvant (Cycles 1-3)	Day 1, 22, 43	Randomization, IV Infusions (Q3W), Safety Assessments, Efficacy tracking
Surgery	Post-Cycle 3	Interval Debulking Surgery, Tumor Tissue Collection
Adjuvant (Cycles 4-6)	Q3W post-surgery	IV Infusions (Q3W), Safety Assessments
End of Study / Follow-Up	Every 12 weeks	Final Vital Signs, Radiographic Imaging, Survival Tracking, IP Accountability

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change From Baseline in Circulating Tumor Deoxyribonucleic Acid (ctDNA) at Cycle 3 (ΔctDNA). Blood samples were collected to determine levels of ctDNA. The fold change in the mean mutant/tumor molecules per mL (MTM/mL) at Cycle 3 from baseline is presented. **Measurement Method:** Central Laboratory Analysis of blood samples for ctDNA quantification.

7.2 Secondary & Exploratory Endpoints * Participants With Surgery and Pathological Complete Response (pCR): Change From Baseline in ctDNA. * Association of Change From Baseline in ctDNA With pCR. * Participants With Surgery and Chemotherapy Response Score (CRS): Change From Baseline in ctDNA. * Progression-Free Survival (PFS). * Overall Survival (OS). * Objective Response Rate (ORR). * Complete Resection Rate during interval debulking surgery.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via investigator assessment, patient reports, and laboratory evaluations throughout the trial. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, including SUSAR (Suspected Unexpected Serious Adverse Reaction) reporting in the EU per Regulation 536/2014. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by the Sponsor's Early Clinical Development Statistics Department.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy (ΔctDNA). Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Target enrollment of $N = 160$, calculated to provide sufficient power to detect a statistically significant reduction in ctDNA levels at Cycle 3 between the experimental and control arms. No multiplicity adjustment is applied. **Handling of Missing Data:** Standard missing data imputation protocols as predefined by the Sponsor's statistical guidelines.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and in compliance with Regulation (EU) 536/2014. **Monitoring Plan:** Regular onsite and remote monitoring visits to assure data integrity and protocol compliance. **Audit Trail:** Formal serious breach reporting, specified record retention, and strict data privacy/protection rules.

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11. Study Identifiers & Governance

Posting ID: {{890374987997240463}} **Primary ID:** {{MK-4830-002}} **Registry Number:** {{NCT05446870}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{Pembrolizumab With Chemotherapy and MK-4830 for Treating Participants With Ovarian Cancer (MK-4830-002)}} **Official Regulatory Title:** {{A Randomized, Phase 2 Study of Pembrolizumab And Chemotherapy With or Without MK-4830 as Neoadjuvant Treatment for High-Grade Serous Ovarian Cancer}} **UMLS Preferred Name:** {{Clinical trial}}

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12. Clinical Framework

Primary Condition: {{High-grade Serous Ovarian Carcinoma, Ovarian Carcinoma}} **Diagnosis Threshold:** {{FIGO Stage III or IV}} **Medical Methodology:** {{Neoadjuvant and Adjuvant Chemo-Immunotherapy}} **Optimization Target:** {{Reduction in circulating tumor DNA (ctDNA) and tumor debulking}}

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13. Study Procedures & Methodology

Management Pathway: {{Standard clinical pathway for HGSOC including Interval Debulking Surgery}} **Education Protocol (HCP):** {{Investigator Protocol Training and Safety Reporting}} **Education Protocol (Patient):** {{Informed Consent, Treatment Expectations, and Surgery Preparation}} **Quality Audit Mechanism:** {{Routine clinical site monitoring and data validation}}

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14. Observation & Follow-Up Schedule

Total Duration: {{Up to disease progression, unacceptable toxicity, or trial completion}} **Onsite Visit Cadence:** {{Day 1 of every 21-day cycle (Q3W) during active treatment}} **Remote Monitoring Frequency:** {{As clinically indicated}} **Checklist Review Interval:** {{Every 12 weeks during the follow-up period}}

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY]*** **| |** ***Lead Statistician*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |